**PROJECT DESIGN PHASE**

# Solution Architecture

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|---|---|
| Date | 27 September 2026 |
| Team ID | 23KD1A1507 |
| Project Name | Automated Network Request Management in ServiceNow |
| Maximum Marks | 4 Marks |

## Solution Architecture:

Solution architecture bridges the gap between business problems and technology solutions. Its goals are to:
• Find the best tech solution to solve existing business problems.
• Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
• Define features, development phases, and solution requirements.
• Provide specifications according to which the solution is defined, managed, and delivered.

## Solution Architecture Diagram

**Automated Network Request Management - Solution Architecture**

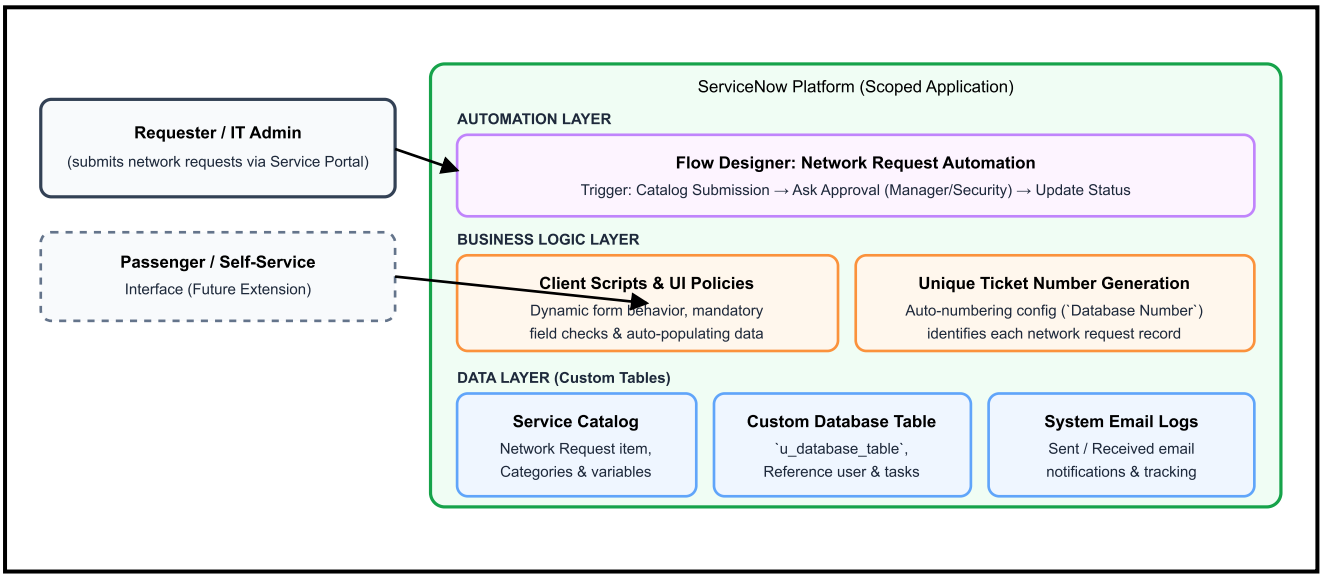



**Figure 1: Architecture and data flow of the Automated Network Request Management System**

### Architecture Components

| Layer | Components | Role |
|---|---|---|
| Data layer | Service Catalog, Custom Database Table (`u_database_table`), and System Email Logs tables | Store network request catalog items, variable definitions, request records (`Database Number`, `Assignment To`, `Mobile Number`, `Type of Connection`, `Approval Status`), and email notification logs. |
| Business logic layer | Client Scripts, UI Policies, and unique ticket auto-numbering configuration | Enforce dynamic form behavior, mandatory field validations, auto-populate requester details, and give each network request its unique tracking identifier. |
| Automation layer & Future extensions | Flow Designer flow "Network Request Automation", Approval Engine, REST integration hubs | Triggered on catalog record creation; routes manager and security approvals, sends automated email notifications, and updates request status. Proposed extensions include hardware switch auto-provisioning and dashboard analytics. |

### Data Flow

- **1.** A request record is created in the custom database table (`u_database_table`) via Service Catalog submission with connection type, customer address, and requester details.
- **2.** Client Scripts and UI Policies validate input fields and auto-populate user information dynamically.
- **3.** Each request is identified by its unique auto-generated ticket number (`Database Number`).
- **4.** The Flow Designer automation workflow runs on record creation, triggers manager/security approvals, updates the approval status, and sends email notifications.

*Note: The exact order of UI policy execution, ticket number generation, and flow triggers depends on their configured platform event order and timing.*

### Implementation Notes

- Tested and verified via ServiceNow Personal Developer Instance (PDI) Service Portal (`/sp`).
- Workflows and approval chains are fully operational and verified through Flow Designer execution logs.
- QR-code generation, hardware provisioning integrations, and external gateway APIs are planned future enhancements.
- The system configuration provides a flexible, scalable architecture for enterprise IT request management.

**Reference:** ServiceNow Developer Portal - https://developer.servicenow.com and ServiceNow Product Documentation - https://www.servicenow.com/docs/